Piyusha Patil

Stockholm, Sweden +46 727683139 ppiyusha2001@gmail.com LinkedIn GitHub Bio

EDUCATION

KTH Royal Institute of Technology **Aug 2026 – June 2028 (Present)**
Master of Science in Computer Science

Savitribai Phule Pune University **Aug 2019 – July 2023**
Bachelor of Engineering in Computer Engineering
CGPA: 9.51/10 (2nd in Department)

RESEARCH/ WORK EXPERIENCE

KTH Royal Institute of Technology **Sep 2026 – Present**
Research Assistant - ASSERT Research Group *Stockholm, Sweden*

- Conducting research under Prof. Martin Monperrus on LLM-based coding agents and inference efficiency, contributing to SuperLean AI with the goal of reducing token consumption by 50% while preserving task performance and user experience.
- Investigating tool debloating and context optimization using Jev, exploring token reduction across tool definitions, descriptions, terminal commands, and agent interactions; evaluating approaches including Headrooms and Ripwire.
- Designing large-scale controlled experiments on SWE-bench Verified (500 real-world tasks), benchmarking agent configurations across 1,000 trials to study token efficiency, task success rates, and reliability using reproducible, isolated execution environments.

KTH AI Society **Oct 2026 – Present**
AI Developer - Project Twiga *Stockholm, Sweden*

- Contributing to Twiga, an open-source, LLM-powered WhatsApp educational assistant for Tanzanian teachers, leveraging Retrieval-Augmented Generation (RAG) to deliver curriculum-grounded responses, lesson planning, and educational content generation. [GitHub]
- Optimizing the RAG retrieval pipeline by improving document chunking and context quality, addressing oversized chunks containing up to 90% of an entire textbook that severely limit retrieval coverage and response accuracy.
- Working on data warehousing and ingestion pipelines, refining mathematical text segmentation, chunk reprocessing, and vector embedding regeneration while investigating automated chunk-size validation and citation integrity across stored conversations.

Indian Institute of Science (IISc), Bangalore **July 2025 – June 2026**
Research Intern – NMCAD Lab, Department of Aerospace Engineering *Bengaluru, Karnataka*

- Working on **Project Urdhyuth under Prof. Dr. Dineshkumar Harursampath**, focusing on conceptual design and software simulation of eVTOL aircraft for Urban Air Mobility (UAM).
- Leading the Machine Learning sub-team to develop intelligent safety algorithms and performing software-in-the-loop (SIL) simulations for avionics subsystems.
- Developed computational thermal models to simulate heat dissipation in embedded electronics and sensor data streams, optimizing cooling strategies without physical hardware interaction - [**GitHub**].
- Leading ML efforts on Battery Management (SOC/SOH prediction), energy-efficient path planning, and predictive maintenance; developed a network reliability model with **94%** precision on synthetic data [**GitHub**].

Protex AI (Y Combinator S21) – Dublin, Ireland **Oct 2022 – Mar 2025**
Software Engineer *Dublin (Remote)*

- Joined at Series A and contributed to product scaling during a high-growth phase leading to Series B funding. Led multiple projects end-to-end, managing architecture, database design, and frontend componentisation to deliver scalable safety intelligence products.
- Developed internal tools using **Retool**, reducing engineering dependency by over **60%** and enabling autonomous operations for non-technical teams.
- Implemented **multi-channel integrations (Slack, Teams, Chime)** for real-time safety alerts, improving incident response time by **40%**.
- Engineered **Auto QA Event Detection** to optimize the internal QA Application, reducing manual efforts by **70%**, and implemented **Email-Based Authentication** to cut failed logins by **16.3%**.
- Resolved a critical pagination bottleneck in a core product feature, cutting load time from **15s to 2s**, and achieved **80% test coverage** with Cypress to ensure system stability.

NOTABLE PROJECTS

Synapse AI - Explainable Brain Age Prediction | *Python, XGBoost, SHAP, Streamlit* **Sep 2026**

- Developed an XGBoost-based brain-age prediction model using 270 structural MRI biomarkers, achieving a mean absolute error (MAE) of 7.15 years and a Pearson correlation of 0.876 on held-out test data, with bias correction and SHAP-based explainability.
- Built an interactive Streamlit application integrating scientific evidence retrieval and LLM-generated explanations to translate MRI predictions into interpretable, personalized insights. [GitHub]

Transformer From Scratch | *Python, PyTorch, Streamlit* **July 2026**

- Implemented a complete Transformer architecture from scratch, reproducing multi-head self-attention, positional encoding, feed-forward networks, residual connections, layer normalization, and encoder-decoder blocks.
- Built an end-to-end English-to-Italian translation pipeline using the OPUS Books dataset, implementing tokenization, batching, training, inference, and greedy decoding in PyTorch.
- Developed an interactive attention visualization tool to analyze attention patterns across encoder-decoder layers and heads, providing interpretable insights into model behavior [**GitHub**].

LLM Code Evaluation Framework | *Python, Streamlit, Hugging Face* **July 2026**

- Developed a modular framework to benchmark LLM-generated code on HumanEval and MBPP using automated execution and functional correctness evaluation.
- Implemented reproducible metrics, failure categorization, and JSON-based evaluation reports for systematic analysis.
- Built an interactive dashboard for cross-model and cross-benchmark comparison[**GitHub**].

HELIOS – Multi-Stage LLM Hallucination Detection | *Python, PyTorch, Hugging Face* **June 2025**

- Designed a multi-stage hallucination detection framework integrating knowledge-graph grounding, semantic validation, web verification, and deliberative reasoning.
- Achieved **0.79 F1** on 1.4K QA pairs using Llama-3.1-8B-Instruct; submitted to ACM TIST Journal.

PUBLICATIONS

[1] **Piyusha Patil***, Dhiaa Mejdi*, Harsh Deshpande*, Sayali Padmawar*, Dr. Aarti Agarkar*. ***HELIOS: An Intelligent Orchestration Framework for Hallucination Detection in Large Language Models.*** Under review at **ACM TIST Journal**. [Confidential manuscript]

[2] **Piyusha Patil**, Harsh Deshpande, Sayali Padmawar, Ashwitha Kassetty, Dr. Aarti Agarkar. ***Bridging Fact and Generation: Comparative Insights into Hallucination Management in LLMs.*** Presented at *2025 IEEE Pune Section International Conference (PuneCon).* [Paper]

[3] Parth Mangalkar, **Piyusha Patil**, Manjusha Katkhede, Prof. Sonal Fatangare. ***CauseVCare – A Blockchain-Based Charity DApp.*** **International Research Journal of Engineering and Technology (IRJET)**, May 2023. [Paper] [Code]

ACHIEVEMENTS & LEADERSHIP

- **KTH Student Ambassador 2026–2027:** Representing KTH and supporting prospective and incoming international students.
- **IEEE PuneCon 2025:** Awarded **Best Paper** in the **Security, Ethics, and Governance** track for *Bridging Fact and Generation: Comparative Insights into Hallucination Management in LLMs.*
- **Spaceport America Cup 2021:** Achieved **33rd overall** rank internationally, **16th in 10,000 ft COTS** Solid Motor category, and **12th** in Project Technical Report among 75 universities worldwide (Official Score Sheet Link).
- **Latin American Space Challenge 2021:** Secured **2nd place** in the 0.5 km Solid Motor Category, ranked **6th overall internationally** and **1st nationally**; featured in state-level newspapers for the achievement (LASC Official Results Link).
- **Top 50 Contributor - GirlScript Summer of Code 2022:** Ranked **44th among 8000+ contributors** nationwide for consistent open-source contributions in web development and IoT.
- **Postman Student Leader:** Served as the official Postman Student Leader for my college, conducted hands-on API literacy workshop for 200+ students and mentored peers on designing and testing RESTful APIs using Postman (Badge Link).

SKILLS

Languages: Python, JavaScript, TypeScript, Solidity, C++
ML/AI: PyTorch, scikit-learn, Hugging Face Transformers, MLOps, Cursor, Claude, LLM evaluation, AI agents
Databases: MongoDB, PostgreSQL, MySQL, Snowflake
Frameworks/Tools: Node.js, React, Redux, Vue, Vuex, Express, MATLAB Simulink, Docker, DBT, AWS, Retool, Git